FREQUENCY & CUMULATIVE FREQUENCY

Relative Frequency • Cumulative Frequency • Cumulative Relative Frequency

BUSINESS ANALYTICS

Understand business data distributions and convert frequency tables into actionable insights.

Learning Objectives

Understand frequency, relative frequency, cumulative frequency and cumulative relative frequency.

Calculate each measure from a business dataset.

Interpret counts, proportions, running totals and cumulative percentages.

Use the measures to answer customer, sales and marketing questions.

Translate statistical summaries into business decisions.

Four Measures at a Glance

Measure	Meaning	Formula	Business question
Frequency (f)	Number of observations in a class/category	Count	How many?
Relative Frequency (RF)	Proportion of observations	f / N	What proportion?
Cumulative Frequency (CF)	Running total of frequencies	Σf up to class	How many up to here?
Cumulative Relative Frequency (CRF)	Running total of relative frequencies	CF / N	What % up to here?

1. Frequency — How Many?

Definition

Frequency is the number of observations or transactions within a category or class interval.

Business question

How many transactions fall within a particular sales-value range?

Frequency (f) = Number of observations in the category

Example

₹1,001–₹1,500 has frequency 200 → 200 transactions.

2. Relative Frequency — What Proportion?

Definition

Relative frequency expresses a class frequency as a proportion of the total.

Business value

It makes categories comparable using proportions or percentages.

$$RF = f / N$$

$$RF\ (\%) = (f / N) \times 100$$

3. Cumulative Frequency — How Many So Far?

Definition

Cumulative frequency is the running total of frequencies through the current class.

Business question

How many transactions are at or below a particular sales threshold?

$$CF = f_1 + f_2 + \ldots + f_i$$

Example

At ₹1,500: 50 + 100 + 200 = 350 transactions.

4. Cumulative Relative Frequency — What % So Far?

Definition

Cumulative relative frequency is the running total of relative frequencies.

Business question

What percentage of transactions are at or below a particular threshold?

$$CRF = CF / N$$

$$CRF\ (\%) = (CF / N) \times 100$$

BUSINESS EXAMPLE

E-COMMERCE SALES TRANSACTIONS

Business Scenario

An e-commerce company has 500 sales transactions.

Management wants to understand the distribution of transaction values.

The company wants to identify the most common sales range.

It also wants to know what percentage of transactions are below selected spending thresholds.

Business objective

Use the distribution to support customer segmentation, promotional thresholds and sales planning.

Complete Frequency Distribution

Sales Amount	Frequency	Relative Frequency	Cumulative Frequency	Cumulative Relative Frequency
₹0–₹500	50	10%	50	10%
₹501–₹1,000	100	20%	150	30%
₹1,001–₹1,500	200	40%	350	70%
₹1,501–₹2,000	100	20%	450	90%
₹2,001+	50	10%	500	100%
Total	500	100%	500	100%

Key observation

₹1,001–₹1,500 is the most frequent class and represents 40% of all transactions.

Step 1: Frequency

For ₹1,001–₹1,500, the frequency is 200.

This is a count, not a percentage.

f = 200 transactions

Interpretation

200 out of 500 transactions are between ₹1,001 and ₹1,500.

Step 2: Relative Frequency

Frequency = 200; total transactions = 500.

Relative frequency converts the count into a proportion of the total.

RF = 200 / 500 = 0.40 = 40%

Interpretation

40% of all transactions are between ₹1,001 and ₹1,500.

Step 3: Cumulative Frequency

Start with the first frequency and keep adding the next frequency.

At ₹1,500: 50 + 100 + 200 = 350.

CF = 50 + 100 + 200 = 350

Interpretation

350 transactions are ₹1,500 or less.

Step 4: Cumulative Relative Frequency

Divide cumulative frequency by the total number of transactions.

At ₹1,500, cumulative frequency = 350 and total = 500.

CRF = 350 / 500 = 0.70 = 70%

Interpretation

70% of all transactions are ₹1,500 or less.

INTERPRETATION

FOUR MEASURES • FOUR BUSINESS QUESTIONS

What Does Each Measure Tell the Business?

Measure	Example result	Interpretation
Frequency	200	200 transactions are ₹1,001–₹1,500.
Relative Frequency	40%	40% of transactions are ₹1,001–₹1,500.
Cumulative Frequency	350	350 transactions are ₹1,500 or less.
Cumulative Relative Frequency	70%	70% of transactions are ₹1,500 or less.

Memory rule

Frequency = COUNT | Relative Frequency = SHARE | Cumulative Frequency = RUNNING COUNT | Cumulative Relative Frequency = RUNNING SHARE

Business Application: Customer Segmentation

≤ ₹500 → 50 transactions (10%).

₹501–₹1,500 → 300 transactions (60%).

> ₹1,500 → 150 transactions (30%).

The company can create different marketing strategies for each segment.

Decision

Higher-value transactions can be targeted with premium offers or loyalty benefits.

Business Application: Promotional Threshold

The cumulative relative frequency at ₹1,500 is 70%.

Therefore, 70% of transactions are at or below ₹1,500.

The company could test ₹1,500 as a promotional or loyalty-program threshold.

Then measure whether the threshold increases average order value.

70% of transactions ≤ ₹1,500

Business Application: Sales Analysis

Question	Measure	Business use
Which range has the most transactions?	Frequency	Identify the most common order value
What share comes from each range?	Relative Frequency	Compare customer segments
How many are below a threshold?	Cumulative Frequency	Set operational/promotional thresholds
What percentage are below a threshold?	Cumulative Relative Frequency	Estimate campaign reach

EXCEL ANALYTICS

BUILD THE TABLE FROM RAW DATA

Excel: Frequency with COUNTIFS

Assume raw transaction amounts are in B2:B501.

To count transactions from ₹1,001 to ₹1,500, use COUNTIFS.

Repeat the formula for each class interval.

```
=COUNTIFS(B2:B501,">=1001",B2:B501,"<=1500")
```

Expected result

200 transactions.

Excel: Relative Frequency

If frequencies are in C2:C6 and the total is in C7, divide each frequency by the total.

Use an absolute reference for the total when copying down.

=C2/C7

Format

Format the result as Percentage.

Excel: Cumulative Frequency

First cumulative frequency = first frequency.

Each next cumulative frequency adds the current frequency to the previous cumulative total.

Copy the formula downward.

First CF = B2

Next CF = Previous CF + Current Frequency

Excel: Cumulative Relative Frequency

Divide each cumulative frequency by the total number of observations.

Format the result as a percentage.

The final cumulative relative frequency should equal 100%.

=Cumulative_Frequency / Total_Frequency

=CF / N × 100

HANDS-ON ACTIVITY

BUSINESS ANALYTICS IN EXCEL

Student Activity

Use a sales dataset containing at least 100 transactions.

Create meaningful sales-value class intervals.

Calculate frequency for every interval.

Calculate relative frequency and percentage.

Calculate cumulative frequency.

Calculate cumulative relative frequency.

Create a suitable chart and write three business insights.

Deliverable

Excel workbook + frequency table + chart + 3 business insights + 2 recommendations

Questions Students Should Answer

Question	Measure
Which sales range is most common?	Frequency
What percentage of sales fall in each range?	Relative Frequency
How many transactions are ≤ ₹1,500?	Cumulative Frequency
What percentage are ≤ ₹1,500?	Cumulative Relative Frequency
Which threshold could be useful for a promotion?	Cumulative Relative Frequency

From Statistics to Business Decision

1. Count observations.
2. Convert counts to proportions or percentages.
3. Calculate running totals.
4. Identify meaningful thresholds.
5. Segment customers or transactions.
6. Design a business action.
7. Measure the result and refine the strategy.

DATA → FREQUENCY → RELATIVE → CUMULATIVE → INSIGHT → DECISION

Quick Check

Q1

1,000 transactions; 250 are electronics. What is the relative frequency?

Q2

Frequencies are 50, 100 and 200. What is the cumulative frequency after class 3?

Q3

350 of 500 transactions are ≤ ₹1,500. What is the cumulative relative frequency?

Q4

What does a cumulative relative frequency of 90% at ₹2,000 mean?

Challenge

How can cumulative relative frequency help a company choose a customer-spending threshold?

Key Takeaways

Frequency = HOW MANY observations.

Relative frequency = WHAT PROPORTION / PERCENTAGE.

Cumulative frequency = HOW MANY UP TO THIS POINT.

Cumulative relative frequency = WHAT PERCENTAGE UP TO THIS POINT.

Together, these measures help businesses understand distributions, segment customers and select useful thresholds.

Remember

Count → Share → Running Count → Running Share

Final Business Insight

Finding 1

40% of transactions are between ₹1,001 and ₹1,500.

Finding 2

70% of transactions are ₹1,500 or less.

Finding 3

30% of transactions are above ₹1,500.

Recommendation

Test ₹1,500 as a promotional or loyalty threshold and monitor its impact on average order value.

Data Analytics by Durga

Frequency tables become valuable when statistical summaries are connected to business decisions.

Business Use Cases: Frequency Analysis

Frequency analysis tells a business how often a category, event or value range occurs.

Retail: number of orders in each order-value range.

Banking: number of transactions in each transaction-value band.

HR: number of employees in each salary or performance range.

Operations: number of days in each demand or production range.

Marketing: number of customers in each purchase-value segment.

Core question

Frequency answers: “How many observations fall into this category or range?”

Use Case 1: Retail & E-Commerce

Order Value	Frequency	Relative Frequency	Business Meaning
₹0–₹500	120	12%	Small-value orders
₹501–₹1,000	250	25%	Low-to-mid orders
₹1,001–₹2,000	400	40%	Most common range
₹2,001–₹5,000	180	18%	Higher-value orders
> ₹5,000	50	5%	Premium orders

Decision

The company can focus upselling and bundles around the most common order range while developing premium strategies for high-value orders.

Use Case 2: Customer Segmentation

Businesses can group customers by spending level.

Frequency shows the number of customers in each spending segment.

Relative frequency shows the percentage contribution of each segment.

Cumulative relative frequency shows the percentage of customers below a spending threshold.

Example: If 70% of customers spend ≤ ₹1,500, ₹1,500 may be tested as a loyalty or promotional threshold.

Analytics value

Turn transaction distributions into actionable customer segments.

Use Case 3: Banking & Financial Services

Transaction Amount	Frequency	Business Question
₹0–₹1,000	500	How many small transactions?
₹1,001–₹5,000	1200	Where is routine activity concentrated?
₹5,001–₹10,000	600	How many medium-value transactions?
> ₹10,000	200	How many high-value transactions?

Applications

Transaction monitoring, customer profiling, service planning and threshold analysis.

Use Case 4: Marketing Campaign Analysis

Suppose a retailer sends a promotion to customers based on purchase value.

Frequency identifies how many customers fall into each purchase band.

Relative frequency identifies the share of customers in each band.

Cumulative relative frequency tells marketers how much of the customer base lies below a selected threshold.

Example: A “Spend ₹2,000 and get a reward” campaign can be evaluated against the existing spending distribution.

Business question

Is the promotion threshold realistic for the majority of customers?

Use Case 5: Human Resources

Monthly Sales / Performance	Frequency	Relative Frequency
Below target	20	20%
Meets target	45	45%
Above target	25	25%
Exceptional	10	10%
Total	100	100%

HR decision

Frequency analysis can support training priorities, incentive design and workforce planning.

Use Case 6: Operations & Demand Planning

An operations team can classify daily demand into ranges.

Frequency tells how many days fall into each demand range.

Relative frequency shows how common each demand level is.

Cumulative frequency helps identify how many days are below a capacity threshold.

Cumulative relative frequency can indicate the percentage of days for which current capacity is sufficient.

Example

If 90% of days have demand ≤ 10,000 units, the business can compare that threshold with its production capacity.

Use Case 7: Inventory Management

Daily Demand	Frequency	Cumulative Frequency
0–100 units	30	30
101–200 units	45	75
201–300 units	20	95
>300 units	5	100

Inventory insight

95% of observed days have demand of 300 units or less. This can inform stock and capacity planning.

Use Case 8: Customer Service

Classify customer-support response or resolution times into ranges.

Frequency shows how many cases fall into each service-time range.

Relative frequency shows the percentage of cases in each range.

Cumulative relative frequency can show the percentage resolved within a target time.

Example

If 85% of cases are resolved within 24 hours, management can compare this with the service-level target.

Use Case 9: Sales Territory Analysis

Monthly Revenue Range	Number of Territories	Relative Frequency
< ₹10 Lakh	8	20%
₹10–₹25 Lakh	14	35%
₹25–₹50 Lakh	12	30%
> ₹50 Lakh	6	15%

Management use

Identify where most territories fall, prioritize coaching, and design differentiated sales targets.

Use Case 10: Quality Control

Manufacturing teams can classify defects per batch.

Frequency identifies how many batches fall into each defect range.

Relative frequency indicates the proportion of batches with each defect level.

Cumulative frequency can show how many batches remain below an acceptable defect threshold.

This supports process improvement and quality monitoring.

Decision

Use the distribution to identify whether defects are concentrated in a small number of batches or spread across production.

Choosing the Right Business Use Case

Business area	Variable to classify	Useful measure
Retail	Order value	Frequency + Relative Frequency
Marketing	Customer spending	Cumulative Relative Frequency
Banking	Transaction amount	Frequency + Cumulative Frequency
HR	Performance / salary	Frequency + Relative Frequency
Operations	Daily demand	Cumulative Frequency
Inventory	Daily demand	Cumulative Relative Frequency
Customer Service	Resolution time	Cumulative Relative Frequency
Quality	Defects per batch	Frequency + Cumulative Frequency

Business Analytics Decision Framework

Step 1: Define the business question.

Step 2: Select the variable and meaningful class intervals.

Step 3: Calculate frequency.

Step 4: Convert to relative frequency.

Step 5: Calculate cumulative frequency and cumulative relative frequency where thresholds matter.

Step 6: Visualize the distribution.

Step 7: Convert the result into a business action.

Framework

Data → Distribution → Pattern → Insight → Business Decision

Classroom Case Study

An e-commerce company has 1,000 customers.

Management wants to introduce a loyalty program.

Use customer purchase value to create frequency classes.

Calculate frequency, relative frequency, cumulative frequency and cumulative relative frequency.

Answer: What percentage of customers spend ≤ ₹2,000?

Recommend a suitable loyalty threshold and justify it using the distribution.

Student task

Prepare the frequency table and give 3 data-driven recommendations.